PRACTICAL METHODS NOTE / APPROXIMATELY 15 MINUTES

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Predicting component shock after a design change

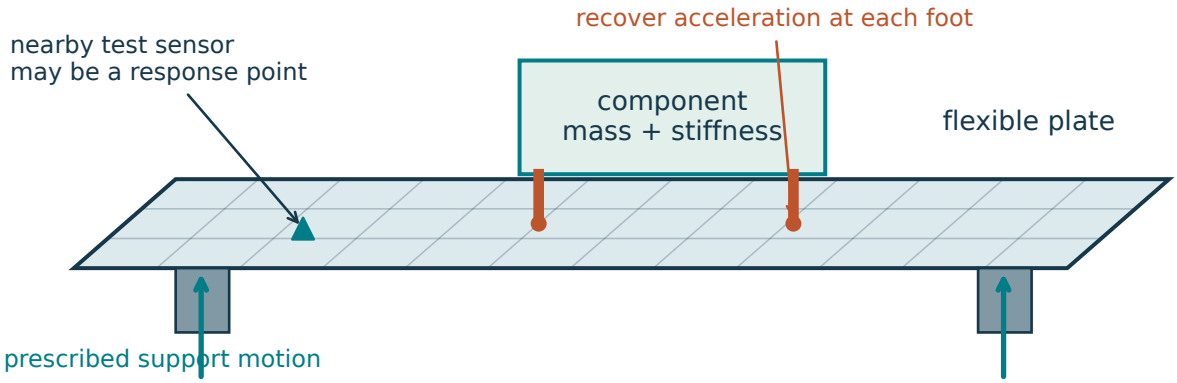


Figure 1. Original conceptual FEM layout. The measurement location, model boundary and component attachment are different physical locations; connecting them correctly is the first analysis decision.

The test campaign is finished. A component is moved, a plate is lightened, or a bracket is added. You have a local shock spectrum and a Nastran model. The defensible question is: what acceleration history reaches the new component attachment, and what SRS does that history produce?

Recommended method: correlate a model of the tested configuration, apply representative shock histories at a justified boundary, run the modified configuration using modal transient analysis (SOL 112) or direct transient analysis (SOL 109), then calculate the output acceleration SRS. SOL 111 can perform the same linear propagation through complex FRFs and inverse Fourier transformation. [1] [2]

If the measured time history is still available, use it. If only its SRS survives, the inverse problem is nonunique: generate several physically credible SRS-compatible inputs, or reconstruct a source on a coupled plate/component model, and report the resulting prediction range. A single matching waveform does not uniquely recover the original event. [3]

This note gives a practical implementation, an illustrative finite-element calculation, and the evidence needed for a post-test design-change assessment. The method is recognized engineering practice; approval to retain qualification depends on the applicable verification plan and the quality of the supporting evidence.

01 / PUT THE MEASUREMENT ON THE CORRECT SIDE OF THE MODEL

A nearby sensor is not automatically a base input

Start with a drawing that identifies the shock source, the measured grid/axis, every plate support, the old and new hardware, and the output coordinates. Specify whether the required output is the environment at the component feet, housing acceleration, internal-board acceleration, or bracket stress. A housing-CG SRS cannot automatically substitute for a mounting-interface qualification spectrum.

For a component carried on a flexible plate, model the plate, bracket, joint stiffness and relevant component mass, CG and inertia together. A rigid mass representation may capture support loading but cannot predict internal component resonances. Use a flexible component model or validated superelement when those modes matter. Avoid rigid spiders that stiffen a large flexible footprint unintentionally.

The best local submodel boundary is an unchanged upstream interface whose motion is measured or credibly reconstructed. Apply the recorded acceleration there. A single point supports a uniform-base assumption only if the interface moves approximately rigidly over the frequency range being used. Different feet can translate and rotate differently; spatial phase and rocking can dominate out-of-plane response. ECSS documents this limitation explicitly. [1]

If the sensor sits on the plate you are modifying, its old acceleration is itself a response of the old coupled system. Prescribing that response to the new model can suppress the very change being investigated. Expand the model upstream, retain the support compliance, and correlate a common source or upstream motion against the old test. If the new mass or stiffness changes the upstream interface motion materially, prescribing the old motion is a conditional assumption, not an established environment.

Record the input provenance: measured event, statistical maximum expected environment, qualification spectrum or envelope of several events. Also preserve axis, units, SRS damping, absolute versus pseudo-acceleration convention, maximax/positive/negative definition, filtering, sampling and usable bandwidth. An envelope may combine peaks that never occurred in one event; a matching synthetic history is a design input rather than recovered telemetry.

For three-axis or multi-support measurements, retain their common time origin. Where relative phase is unknown, evaluate justified coherence/phase scenarios. Driving all feet with the same trace imposes perfect coherence; randomizing them without evidence is equally consequential.

02 / YOUR SOL 111 EXPERIENCE STILL APPLIES

Transfer the waveform, then take the maxima

For a linear model, obtain the complex acceleration-to-acceleration FRF from each imposed support coordinate to each output. Include component mass and the actual support constraints. At Fourier frequency f,

$$A_o(f) = \sum_j H_{oj}(f) A_{b,j}(f)$$

$$a_o(t) = \mathcal{F}^{-1}\{A_o(f)\}$$

Here A denotes a complex Fourier transform, b a base input, and o the output. Retain phase; use the same Fourier scaling, a compatible frequency grid and conjugate symmetry. Extend the record with zeros for physical ringdown and to prevent circular wraparound. Resolve narrow resonances with sufficient FRF frequency spacing. A force-to-acceleration FRF, in g/lbf, is not the dimensionless base-acceleration transfer required here.

Feed the calculated **absolute output acceleration** into an SRS oscillator bank. For each oscillator natural frequency f_s, with $\omega_s = 2\pi f_s$ and relative displacement z,

$$\ddot{z} + 2\zeta_s \omega_s \dot{z} + \omega_s^2 z = -a_o(t)$$

$$S_o(f_s, Q_s) = \max_t |a_o(t) + \ddot{z}(t)| \qquad Q_s = \frac{1}{2\zeta_s}$$

The SRS frequency is the frequency of a hypothetical oscillator, rather than a Fourier bin containing an input amplitude. The maximum operation discards timing, so generally $S_o(f) \neq |H(f)| S_b(f)$. Uniform amplitude scaling of a fixed waveform is valid for a linear system; pointwise spectral scaling is a different operation.

Numerical counterexample. A 100 g, 0.5 ms half-sine drives a 300 Hz structural oscillator with 2% damping. With Q = 10 for both SRS calculations, the input SRS at 300 Hz is 54.7 g and the output SRS is 321.2 g. Multiplying by the steady-state gain 25.02 predicts 1,368.9 g. The actual output time-history peak is only 57.0 g. These are three different quantities. The product overpredicts this example; it is not a general conservative bound.

Original numerical illustration, independently checked against continuous-time state-space integration; no flight or qualification data are represented.

03 / A PRACTICAL NASTRAN IMPLEMENTATION

Make SOL 112 the default starting point

Use SOL 103 to inspect the fixed-interface modes, then SOL 112 with enforced support acceleration. Use SOL 109 as a direct linear comparison where affordable. Contact opening, joint slip or isolator travel limits require an appropriate nonlinear transient treatment. The following mapping is based on MSC Nastran 2022.2; verify syntax and defaults for the installed MSC or Simcenter release. [2]

Purpose	Typical implementation
Modes and damping	METHOD / EIGRL; SDAMPING / TABDMP1 with a documented damping convention. Physical structural damping is separate from SRS oscillator Q.
Prescribed acceleration	SPC/SPC1 identifies support DOFs; SPCD supplies spatial factors; TLOAD1 with TYPE=ACCE references the time table. Select via DLOAD.
Integration and output	TSTEP controls step/count/output interval. Request ACCELERATION at every required point/axis; recover stress and joint force separately.
Modal completeness	Use enforced-motion residual/modal-augmentation vectors as supported by the solver. MSC guidance specifies PARAM,RESVEC,YES. Verify convergence.

Illustrative load linkage; geometry, remaining constraints, modes, damping, TSTEP and the actual data table must be supplied:

```
SPC1,10,3,101
SPCD,30,101,3,1.0
TLOAD1,20,30,,ACCE,40
$ Case control: SPC=10 and DLOAD=20
$ TABLED1 40: time versus acceleration in model units
```

Grid 101/component 3 is an example driven DOF, not a prescription to constrain every support identically. Convert g to model acceleration: 386.09 in/s² per g in an inch-second model. Use consistent mass units or the verified solver mass-scaling convention; lbm is not directly lbf·s²/in.

Recover absolute acceleration once. Some relative-coordinate formulations require adding base motion; other output paths already include it. Check a nearly rigid model: its absolute output must reproduce the prescribed base motion. Confirm the solver actually applied acceleration, not force. This small benchmark detects sign, unit and double-addition errors quickly.

MSC also supports SRS generation inside SOL 109/112 using RSPECTRA, DTI selection and spectral output commands. External validated SRS postprocessing is often easier to audit. Spectrum *application* in SOL 103 instead combines modal peak estimates; it does not directly produce an output-location SRS. [2]

04 / AN ILLUSTRATIVE FINITE-ELEMENT DESIGN CHANGE

A lower time-history peak can hide a larger SRS

Model a cantilever strip bracket with 12 Euler-Bernoulli beam elements, consistent mass and a 0.25 lbm translational tip mass. Length is 4.00 in and width 1.50 in. Change thickness from 0.125 to 0.100 in. Use E = 10.15 Mpsi, density = 0.0975 lbm/in³ and 2% modal damping. Impose the same 100 g, 0.5 ms half-sine acceleration at the root; recover absolute tip acceleration and its Q = 10 maximax SRS.

Solve the eigenproblem and modal equations, where Γ_r is base-motion participation and q_r is the modal coordinate:

$$\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2 \mathbf{M}\boldsymbol{\phi}_r \qquad \ddot{q}_r + 2\zeta_r\omega_r\dot{q}_r + \omega_r^2 q_r = -\Gamma_r a_b(t)$$

This example was solved independently in Python, not run in Nastran. It uses all 24 elastic modes of each beam model, a 2.5 µs step and 0.5 s record; the oscillator recurrence uses a piecewise-linear input. The tip mass has no rotary inertia in this teaching model. It is a bracket idealization, not a validated shell model of your hardware.

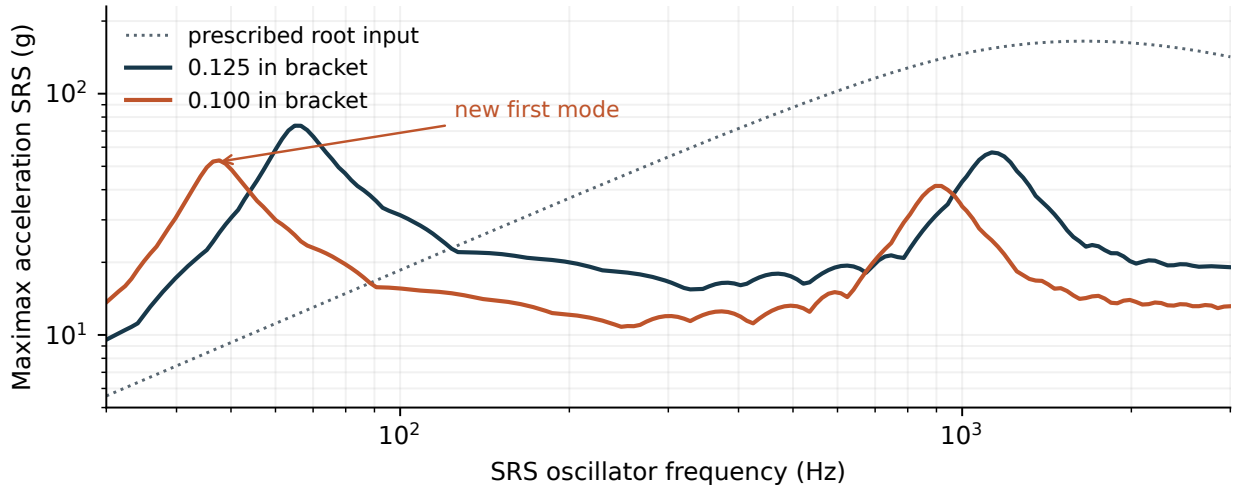


Figure 2. Original FEM prediction. Every curve uses the same SRS definition. The new first-mode peak shifts toward 47 Hz.

Quantity	0.125 in	0.100 in
First / second mode	65.2 / 1128.6 Hz	47.0 / 899.9 Hz
Peak tip acceleration	18.1 g	12.7 g
SRS at 46.46 Hz	24.4 g	52.3 g
SRS at 895.33 Hz	31.1 g	41.5 g

Thinning reduces the physical peak yet more than doubles SRS near the new first mode. Mesh refinement from 12 to 24 elements changes SRS by less than 0.02% on the evaluated 30-3000 Hz grid; halving the 5 µs step changes it by less than 0.03%. Those checks establish numerical consistency of this idealization, not hardware correlation.

05 / WHEN THE ONLY INPUT LEFT IS AN SRS

Use a conditional prediction with visible assumptions

First try to recover the original records. SRS alone does not specify duration, phase, support-to-support timing or frequency-dependent arrival times. NASA has documented why matching spectra can conceal materially different waveforms. [3]

A practical SRS-only analysis begins with the target definition and available event knowledge. Synthesize several candidate accelerations using wavelets, damped sinusoids, or modified representative records. Match the chosen SRS convention across the full band, then also check duration, rise/decay, peak acceleration, velocity change, displacement drift and usable Fourier content. Enforce zero net velocity or displacement only when the event physics warrants it; a half-sine pulse, for example, has a nonzero velocity increment.

Propagate every candidate through both old and modified models. Vary physical damping, uncertain joints, input directions and plausible inter-support relationships. Envelope or summarize the resulting output SRS according to the agreed uncertainty method. A handful of synthetic realizations is a sensitivity study; it does not establish a 95/50 statistical environment or a mathematical worst case.

Where plate flexibility governs the mounting field, use a **coupled plate/component transient model**. Fit source or upstream forcing to baseline measurements, then modify the hardware with that source retained. ECSS discusses pulse tuning and approximate correction for interface-spectrum mismatch; NASA's TFE work provides another source/structure formulation. Both require a credible source and support field. [1] [5]

There are also established screening approaches. ECSS describes empirical FRF-to-shock-transmissibility procedures for defined electronic-unit conditions, and SRS-ratio extrapolation with limitations. These are scoped approximations, not a universal equality between a sinusoidal FRF and an SRS ratio. Their frequency regions and margins should not be transferred automatically to an arbitrary plate or bracket. [1]

If baseline response data exist at the relevant location, an engineering update can be expressed as

$$\widehat{S}_{\text{new}}(f_s) = S_{\text{old, test}}(f_s)\,\frac{S_{\text{new, model}}(f_s)}{S_{\text{old, model}}(f_s)}$$

Both model spectra must come from the same defined input realization and SRS settings. Check the ratio across input families and uncertain models, broaden shifted resonances, and avoid divisions by tiny spectral valleys. This calibrated ratio assumes baseline model bias transfers to the new design; present it as an engineering extrapolation with supporting evidence.

06 / WHAT MAKES THE RESULT REVIEWABLE AFTER TESTING ENDS

Validate the old configuration before trusting the new one

Reproduce the as-tested assembly, including flight hardware versus mass simulators, bolt interfaces, component position and sensor locations. Compare predicted and measured baseline time histories, SRS, dominant frequencies and decay where those records exist. Matching only the imposed input is not validation of transmission through the plate. If you have no measured output downstream of the boundary, state that the transfer prediction is unvalidated and use evidence from relevant modal, FRF or heritage tests to constrain it.

Converge the actual output SRS with respect to mesh, modal cutoff, residual treatment, time step, output sampling and response duration. A 10 kHz requested SRS does not mean that extracting modes to exactly 10 kHz suffices. The SRS oscillator bank, structural response and input bandwidth impose different resolution requirements. A practical initial choice is at least 20 samples per shortest relevant period, followed by step halving; this is an analyst starting point, not an acceptance standard. Interpolation cannot restore missing measured bandwidth.

For wave-dominated response, resolve the shortest relevant bending, shear or longitudinal wavelength and refine around joints and discontinuities. A modal model correlated only at a few low frequencies cannot establish high-frequency pyroshock accuracy. Declare a supported prediction band and use measured/heritage bounds or a justified higher-frequency method beyond it. Do not extend an attractive FEM attenuation curve indefinitely.

Compare the modified interface SRS with the component’s demonstrated or approved allowable environment using matching axes, Q, frequency range and mounting conditions. Separately evaluate bracket stress, joint loads, displacement/clearance, isolator travel and any functional shock sensitivity. A single peak acceleration compared with an SRS ordinate is not a like-for-like comparison.

Document the chain from measured environment to nominal prediction, model discrepancy allowance, waveform/support uncertainty and qualification margin. Keep these factors distinct to avoid double counting a margin already embedded in the input. Use broader frequency envelopes when uncertain joint stiffness can shift resonances; a narrow nominal notch is weak evidence for an environment reduction.

The final change package should identify the retained test evidence, modified configuration, input assumptions, correlation results, sensitivity envelope, predicted component-interface SRS and disposition against its allowable. If the change introduces a new load path, contact regime or unsupported frequency region, a focused local test may be the evidence needed even when the original system environment campaign is complete.

NASA-STD-7003A supplies pyroshock test criteria, rather than blanket analysis-only qualification authority. This note cites the verified 2011 edition; apply your program’s controlling revision and tailoring. ECSS-E-HB-32-25A is guidance. [4]

07 / METHOD SELECTION AND REFERENCES

The recommendation for this modification

Build the old and new plate/bracket/component assemblies. If the plate-support time history you mentioned is available and that boundary remains representative, drive it directly in SOL 112, or use SOL 111 complex transfer functions with FFT/IFFT. Recover acceleration at each new attachment and compute its SRS. Correlate the old configuration, then carry the demonstrated discrepancy and the change-specific uncertainty into the new environment.

If only a local SRS is available, use the same framework with multiple physically credible matched histories or a calibrated coupled source model. Report the answer as conditional on those inputs. Use ordinary response-spectrum analysis primarily for approximate peak stress/force screening, and use empirical spectral scaling only within its demonstrated applicability.

Three points to retain

1. Choose a boundary whose motion remains valid after the change.
2. Propagate the acceleration waveform before calculating the output SRS.
3. Correlation and uncertainty determine whether the prediction can support the qualification disposition.

Reflection: the new bracket lowers the output peak g. Has the component shock environment improved? **Not necessarily.** The worked FEM lowers peak g from 18.1 to 12.7 while raising SRS near 47 Hz from 24.4 to 52.3 g. Compare the spectrum and relevant failure response over the full supported band.

Primary and official references

[1] ECSS-E-HB-32-25A, Mechanical shock design and verification handbook, 14 July 2015. Sections 8.2.4, 9.1-9.2.4 and 18.2.3-18.2.4; printed pp. 81-83, 129-149 and 467-479. Handbook guidance. Official listing; full report also available from the linked mirror below.

[2] Hexagon, MSC Nastran 2022.2 Dynamic Analysis User's Guide. Chapter 7, pp. 295-300: enforced motion and residual vectors. Chapter 9, pp. 358-369: spectrum application versus generation. Implementation reference for the named release.

[3] R. C. Ferebee et al., An Alternative Method of Specifying Shock Test Criteria, NASA/TM-2008-215253, April 2008. Discusses nonunique SRS-compatible waveforms and information omitted by SRS-only specifications.

[4] NASA-STD-7003A, Pyroshock Test Criteria, 20 December 2011. Sections 1.1-1.4, 4.4, 4.6-4.7 and Appendix A. This is the verified edition used here; the currently controlling program revision must be confirmed.

[5] NASA NESC, New Transient Finite Energy Shock Prediction Methodology, 27 April 2023. Force-based prediction using finite-energy input reconstruction and structural transfer accelerations; useful context for source/structure modelling.

Full ECSS handbook mirror used to inspect the cited pages when the official download was unavailable. All diagrams and numerical examples in this note are original illustrations. No user hardware, measured telemetry or Nastran solver execution is claimed. Prepared using GPT-6 Astra.